



Sign Language Translation for Indian Sign Language(ISL)

Aritra Ambudh Dutta

Mentor: Professor Ashutosh Modi

Abstract

This project explores the application of recent machine learning frameworks to the problem of Indian Sign Language (ISL) translation at the word level. The work is carried out under the guidance of Prof. Ashutosh Modi at the Department of Computer Science and Engineering, IIT Kanpur. The project integrates Optimal Transport Guided Procedure Learning (OPEL) with Indian Sign Language data to enable soft alignment of gesture sequences, allowing for flexibility in temporal variation and non-monotonic sign structure. A major contribution of this work is the curation of a large-scale ISL dataset consisting of approximately 1.5 lakh word-level video clips scraped and cleaned from public sources such as YouTube.

The first phase of the project involved studying the mathematical foundations of Optimal Transport and implementing the OPEL framework for ISL gesture modeling. This included incorporating temporal priors, background frame handling, and contrastive regularization into the alignment mechanism. In parallel, I began integrating SHuBERT, a self-supervised multi-stream transformer model for sign language understanding. SHuBERT extracts contextual representations from face, hand, and body pose streams using masked cluster prediction. While OPEL has been successfully implemented for ISL gesture alignment, the integration of SHuBERT is ongoing. The goal is to develop a modular architecture where SHuBERT encodes rich visual features, and OPEL provides alignment supervision for improved gesture-to-text translation. Together, these frameworks aim to build a more interpretable and generalizable model for ISL, advancing the accessibility and technological support for the Deaf community in India.

Keywords— Indian Sign Language, Optimal Transport, OPEL, SHuBERT, Gesture Alignment

1 Introduction

Sign Language Translation (SLT) is a critical task for improving accessibility for Deaf and hard-of-hearing individuals. Indian Sign Language (ISL), which is used by millions across the country, remains underrepresented in both datasets and research infrastructure. This project focuses on ISL word-level translation using two cutting-edge frameworks: OPEL (Optimal Transport Guided Procedure Learning) and SHuBERT, a self-supervised visual representation model.

The unique challenge in sign language is modeling multimodal and temporally variant gestures. Traditional methods often struggle with rigid alignment or insufficient contextual modeling. OPEL addresses this by treating video sequences as distributions and aligning them flexibly via Optimal Transport. SHuBERT, on the other hand, captures context using multi-stream visual features (hands, face, pose) and trains using masked cluster prediction. This report documents the dataset curation, theoretical grounding, implementation of OPEL, and progress on integrating SHuBERT.

2 Background

2.1 Sign Language and the Need for Translation

Sign languages are complex visual-spatial languages that use hand gestures, facial expressions, and body postures to convey meaning. Unlike spoken languages, they are inherently multimodal and do not follow a linear or phoneme-based structure. This makes sign language translation (SLT) a fundamentally different and more challenging task in natural language processing and computer vision.

Indian Sign Language (ISL), used by millions across the Indian subcontinent, remains under-researched in terms of computational resources and model development. While automatic speech recognition (ASR) and text translation systems have seen significant progress, ISL lacks large-scale datasets, benchmark models, and language tools. This project seeks to address this gap by leveraging recent advances in alignment and representation learning.

2.2 Optimal Transport and the OPEL Framework

Traditional SLT systems often rely on hard, monotonic alignment between gesture sequences and text labels. However, this approach fails in real-world scenarios where the duration of signs varies significantly between signers or even for the same signer across contexts. To address this, the OPEL framework proposes a flexible alignment strategy based on **Optimal Transport (OT)**.

OT models each sequence as a distribution and computes a transport plan that aligns them in a "soft" manner. Instead of enforcing one-to-one matching between frames and tokens, OPEL minimizes a cost function that includes transport costs and temporal regularization. This makes it particularly well-suited for procedural or step-wise tasks like ISL word modeling.

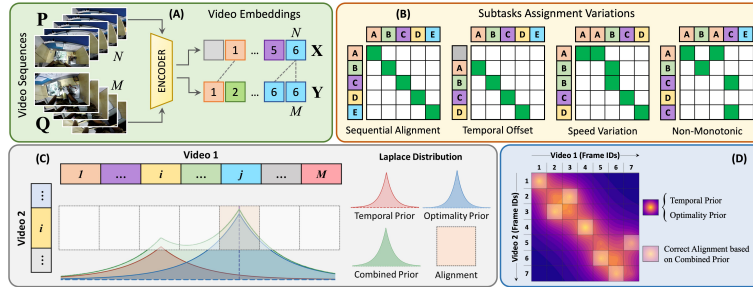


Figure 1: Frame Alignment using OPEL

OPEL Framework: [1] (A) The encoder generates frame-wise embeddings from videos, facilitating subsequent OT calculations. (B) Pair-wise scenarios captured through the assignment matrix- from strictly synchronized actions to temporal shifts and differing action speeds, to non-monotonicity. (C) 1-D depiction of alignment of a single frame (i -th) of Video 2 with its best match frame (j -th) of Video 1, based on the proposed priors. (D) 2-D representation of the optimal alignment of frame sequences.

2.3 SHuBERT and Self-Supervised Representation Learning

While OPEL addresses the alignment problem, it does not explicitly learn deep representations of sign gestures. To enhance the quality of visual features, we incorporate the SHuBERT framework — a self-supervised model for sign language that learns contextual embeddings without labeled data.

SHuBERT uses **multi-stream input** (left hand, right hand, face, and body pose) extracted from each video frame. It applies a **masked cluster prediction** strategy similar to BERT: after clustering the features using k-means,

it randomly masks parts of the sequence and trains a transformer to predict the missing cluster labels. This forces the model to capture temporal structure and inter-modal relationships in an unsupervised way.

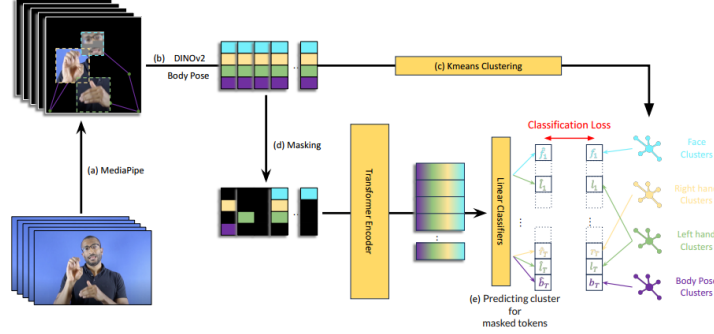


Figure 2: SHuBERT Architecture Overview

SHuBERT Pre-Training Pipeline:[2] (a) Landmarks for face, hands, and body pose are extracted from each video frame using MediaPipe[3], with interpolation applied to fill missing keypoints. The hand and face regions are cropped, while the face is partially masked and blurred for privacy. (b) Visual features are computed for these cropped regions using DINOv2[4], resulting in a four-stream representation for each frame—left hand, right hand, face, and upper body pose. (c) Each stream’s feature vector is mapped to a cluster index using pre-trained k-means clustering. (d) A subset of these feature tokens is randomly masked across time and modality. (e) A transformer encoder is trained to predict the missing cluster assignments, enabling the model to learn contextual sign representations in a self-supervised manner.

Together, OPEL and SHuBERT offer a powerful combination — flexible alignment and robust representation — to tackle the complexities of Indian Sign Language translation.

3 Dataset Construction

3.1 Video Collection and Curation

A critical component of this project involved the creation of a large-scale, high-quality dataset of Indian Sign Language (ISL) word-level gestures. To this end, we developed a custom video scraping pipeline to extract publicly available ISL content from YouTube and similar platforms. These videos primarily consisted of educational content, tutorials, and dictionary-style sign demonstrations.

The raw data was then subjected to a multi-stage manual and semi-automated filtering process. Each video was examined for:

- **Linguistic Relevance:** Ensuring that the sign performed matches a valid and commonly used ISL word or expression.
- **Visual Clarity:** Discarding videos with poor lighting, occlusions, or low frame rates.
- **Signer Diversity:** Preserving videos from various signers across different genders, clothing styles, and backgrounds to enhance generalization.

This resulted in a curated corpus of approximately **150,000** word-level ISL gesture clips, making it the largest known public dataset of its kind.



3.2 Preprocessing and Feature Extraction

To prepare the dataset for use in machine learning pipelines, each video was passed through a structured preprocessing pipeline with the following components:

- **Person Detection and Tracking:** We used YOLOv11 to detect and consistently track the signer across all frames, ensuring that irrelevant background movements were ignored.
- **Landmark Detection:** MediaPipe’s holistic model was applied to extract 3D keypoints corresponding to hands, face, and upper body pose. These landmarks were crucial for forming the multi-stream input used by SHuBERT.
- **Visual Embedding:** Cropped hand and face regions were encoded using DINOv2, a transformer-based self-supervised model. These embeddings provided high-level, transferable visual features suitable for downstream clustering and transformer-based learning.
- **Temporal Alignment and Padding:** To standardize input lengths for model training, sequences were padded or trimmed to fixed frame sizes while maintaining gesture completeness.

This preprocessing framework ensured that each video was clean, structured, and representation-ready — enabling smooth integration with both the OPEL and SHuBERT modeling pipelines.

4 Implementation

4.1 OPEL Framework for Gesture Alignment

The OPEL (Optimal Transport Guided Procedure Learning)[1] framework was implemented to address the alignment challenge in ISL word-level translation. Unlike conventional alignment strategies that assume a strict one-to-one frame correspondence, OPEL models the video frames as samples from probability distributions and computes a soft alignment using Optimal Transport.

- **Optimal Transport Loss:** We used the Sinkhorn algorithm to compute entropy-regularized optimal transport plans between gesture frame embeddings and target textual tokens.
- **Temporal Priors:** Laplace priors were applied to enforce temporal smoothness and discourage unreasonable long-range matches across the frame timeline.
- **Virtual Frames:** To handle background or idle gestures that do not correspond to any semantic content, we introduced virtual background frames. These act as sink nodes in the transport plan, absorbing non-informative visual data.
- **Contrastive Loss:** An inter-video contrastive loss was included to encourage the alignment module to distinguish between different sign classes and signer variations.
- **Encoder Architecture:** The video frames were encoded using a lightweight CNN (ResNet-18 backbone) into a latent feature space, on which the optimal transport and alignment operations were performed.

4.2 SHuBERT Integration Pipeline (Ongoing)

The SHuBERT[2] framework, a self-supervised transformer-based model designed for sign language, was explored for its ability to learn contextual visual features from multiple input streams. We began building the SHuBERT-style pretraining pipeline as follows:



- **Multi-Stream Feature Extraction:** Using MediaPipe[3], we extracted four visual streams — face, body pose, left hand, and right hand — for each frame in the video sequence. Hands and face regions were cropped and encoded using DINOv2[4], a state-of-the-art self-supervised visual representation model.
- **Clustering:** Feature vectors from all four streams were clustered using k-means to obtain pseudo-labels for each modality. These cluster IDs serve as the ground truth for masked prediction.
- **Masked Cluster Prediction:** A transformer encoder was trained to predict the cluster assignments for randomly masked segments of the visual input, enforcing temporal and cross-modal consistency in representation learning.
- **Planned Integration:** The SHuBERT encoder is currently being integrated into the OPEL-based alignment framework. The goal is to use SHuBERT for contextual embedding of the gesture sequence and guide its alignment using OPEL’s soft transport mechanism.

5 Results

- **Dataset:** We curated and preprocessed a large-scale Indian Sign Language dataset consisting of approximately 150,000 word-level videos. This dataset was constructed by scraping publicly available ISL content from YouTube, followed by a multi-stage filtering process to ensure clarity, linguistic consistency, and diversity in signer representation. Each video clip corresponds to a distinct ISL word, making this the most extensive publicly available word-level ISL dataset to date. The dataset will serve as a valuable resource for training and evaluating future models in ISL recognition and translation.
- **OPEL Implementation:** The OPEL framework was successfully adapted for ISL gesture alignment. Using optimal transport, we implemented soft alignment between gesture sequences and textual labels, which allows for flexible matching of signs with varying temporal lengths. To enhance the robustness of alignment, temporal priors (e.g., Laplace distributions) were applied, encouraging smooth transitions across frames. Additionally, virtual background frames were introduced to handle non-informative segments in the input. We also incorporated an inter-video contrastive loss to improve the model’s ability to generalize across different signers and environments.
- **SHuBERT Integration (In Progress):** We extracted multi-stream visual features — including landmarks for face, body pose, left and right hands — using MediaPipe. These features were encoded using DINOv2 and clustered using k-means to generate pseudo-labels for self-supervised learning. We initiated the SHuBERT pretraining pipeline using masked cluster prediction and are currently integrating it into the OPEL-based alignment framework. This integration aims to combine SHuBERT’s contextual representation learning with OPEL’s alignment capabilities, forming a unified model for accurate and interpretable ISL translation.

6 Future Work

While significant progress has been made in implementing the OPEL framework and preparing the SHuBERT integration pipeline, several key developments are planned for the next phase of the project:

- **Complete SHuBERT Integration:** The primary focus will be to fully integrate SHuBERT’s multi-stream transformer encoder into the OPEL-based alignment framework. This includes harmonizing the data flows, ensuring compatibility between embedding formats, and jointly training both components for improved gesture-to-text mapping.
- **Pretraining and Fine-tuning:** SHuBERT will first be pretrained on publicly available, large-scale sign language datasets such as American Sign Language (ASL) corpora to learn generalized sign features. This pretrained model will then be fine-tuned specifically on the curated ISL dataset to adapt to the linguistic and cultural nuances of Indian Sign Language.



- **Quantitative Evaluation:** We aim to evaluate the combined model on benchmark ISL word recognition tasks using metrics such as accuracy, alignment quality, and embedding consistency. Comparative studies against traditional alignment-based baselines will be conducted to assess improvements.
- **Real-Time Deployment:** In the final stage, we plan to explore real-time deployment possibilities by connecting the model to webcam input. This involves optimizing inference speed, developing a user interface, and testing the system with real-time ISL gestures to assess its practical viability.

These next steps will help transition the project from experimental implementation to a scalable, real-world ISL translation system that could support accessibility technologies for the Deaf community in India.

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